Absorption Order and Forcing Spectrum in Erdős–Straus Families

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October 2026

1 Abstract

The Erdős–Straus conjecture asks whether $4/n = 1/x + 1/y + 1/z$ is solvable for every $n \geq 2$ [1]. Recent computational work in this program reduced the problem to two finite data layers: a covering census of 12,827 covering families [12] and a union-forcing census of 78 union groups [12]. This paper studies the order theory and per-cell structure of that union layer.

On the 78 union groups $G = (C, p)$ ($C \subseteq \mathbb{Z}/35$ nonempty, $p \in \{0, 1\}$) we define the *absorption order* $G \preceq H \iff C_G \subseteq C_H \wedge (p_G \Rightarrow p_H)$. We prove [PROVED] that $\preceq$ is a partial order (A1), that it characterizes datum-equality of unions (A2), and that dominated families can be pruned from any union without changing the datum, so minimal covering families are $\preceq$-antichains (A3). For the redundancy rank $\rho(S)$ (maximum removable members under datum-preserving reduction) we prove [PROVED] disjoint superadditivity (R3′), a general corollary (R3″), the overlap-deficit bound (T), and the absorption-collapse formula with uniqueness (R4′) — the naive stated superadditivity being *false* for overlapping families, with exact counterexample, repaired after hostile review. We prove [PROVED] the singleton-factorization theorem: every minimal u-cover of $\mathbb{Z}/35$ splits uniquely into a hard core T times an independent per-cell singleton fill; the census gives 212,264 cores and exactly 7,052,450 minimal u-covers [COMPUTED]×2.

We define the *ES forcing spectrum*: the per-cell 6-class profile $\sigma_S(c) \in \{\mathrm{H}, \mathrm{C}, \mathrm{Fh}, \mathrm{Fo}, \mathrm{Vh}, \mathrm{Vo}\}$ of an instance S. We prove [PROVED] the repaired forced-cell characterization S2(a)′ (the statement as originally written fails on void cells; repaired after review), the pc-secured/void/split characterizations S2(b–d), the six-class partition S3, and union monotonicity S4 ($\psi_{S \cup T} = \max$, with pc-secured wins). The spectrum census (3,081 pair types, 82,272,378 weighted pairs) yields aggregate masses and a complete upgrade anatomy [COMPUTED]×2. We also formalize [PROVED] the two-sided forcing decomposition A–D (CRT, scaling, filter, cardinality) under four documented review conditions.

All claims are [PROVED] by Lean 4.34.1 machine check except the surgical down-set A4, which is explicitly [OPEN]. The formalization note (§3.2) names the modules and theorems.

2 Introduction

The Erdős–Straus conjecture [1] asserts that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers for every $n \geq 2$. It remains open. A long line of work attacks (1) through residue-class covering: fix the modulus 35 and classify the k-classes of $n = 24k + 1$ by which residues they cover and whether they cover primes (Theorems A–D, §4.5).

This program's recent computational work produced two finite data layers, cited here as given [12]: the R281 covering census (12,827 families with covered sets, prime-covering flags, and the surgical classification) and the R285 union-forcing census (78 union groups $G = (C, p)$

with $C \subseteq \mathbb{Z}/35$ nonempty and $p \in \{0,1\}$, 3,081 pair types). This paper is the first theory layer built on the union census: the order structure of the 78 groups and the per-cell forcing profile of an instance.

Our contributions, all [PROVED] unless labeled otherwise:

1. The *absorption order* $\preceq$ on union groups, with A1 (partial order), A2 (absorption characterization: $\text{datum}(G \cup H) = \text{datum}(H) \iff G \preceq H$), and A3 (dominance pruning; minimal covering families are $\preceq$-antichains). Theorem 1–3.

2. The redundancy rank ρ with disjoint superadditivity R3′, general corollary R3″, the overlap-deficit bound T, and repaired absorption-collapse R4′ (Theorems 4–7). The naive stated superadditivity is *false* for overlapping families (Proposition 1, exact counterexample), which we state honestly: the disjoint restriction is the whole theorem.

3. The *singleton-factorization theorem* (Theorem 8): every minimal u-cover of $\mathbb{Z}/35$ factors uniquely as hard core T times per-cell singleton fill, yielding the exact census 7,052,450 [COMPUTED]×2 (Figure 1, Table 2).

4. The *ES forcing spectrum*, a per-cell 6-class profile, with S2(a)′ (repaired forced-cell characterization), S2(b–d), the partition S3, and union monotonicity S4 (Theorems 9–12), plus its 3,081-pair-type census (Figure 2).

5. The two-sided forcing decomposition A–D [PROVED] under four documented review conditions (§4.5).

The honest history is part of the record: two statements as originally written were objected to by hostile review (R3's superadditivity, S2(a)) and both were repaired before formalization; the surgical down-set A4 remains [OPEN] and is not claimed. The formalization note (§3.2) gives the exact Lean theorem names.

3 Methods

3.1 Definitions

We work on the universe $U = \{0, \ldots, 34\}$ of k-classes mod 35, with the six hard cells

$$\text{Hard6} = \{0, 5, 7, 12, 15, 22\}. \tag{2}$$

Definition 1 (Union group and absorption order)**.** A *union group* is $G = (C, p)$ with $\emptyset \neq C \subseteq \mathbb{Z}/35$ and $p \in \{0,1\}$ (the prime-covering flag). The *absorption order* is

$$G \preceq H \quad \iff \quad C_G \subseteq C_H \ \wedge \ (p_G \Rightarrow p_H). \tag{3}$$

For a finite set S of groups the *union datum* is

$$u(S) = \bigcup_{G \in S} C_G, \qquad P_{\text{cov}}(S) = \bigcup_{G \in S,\, p_G = 1} C_G, \tag{4}$$

with the derived quantities $\text{cap}_2 = |u|$, $\text{cap}_1 = |P_{\text{cov}}|$, and $m = |P_{\text{cov}} \cap \text{Hard6}|$ [12].

Definition 2 (B1-redundancy and redundancy rank)**.** $G \in S$ is *B1-redundant* in S iff removing it preserves the union datum [5]. A *reduction step* drops a B1-redundant member; a *core* of S is a terminal B1-minimal set of a maximal reduction sequence. The *redundancy rank* is

$$\rho(S) = \text{max number of removable members} = |S| - \min\{|C| : C \text{ a core of } S\}. \tag{5}$$

The max-removals convention is load-bearing: cores need not be equicardinal [6].

Definition 3 (ES forcing spectrum)**.** For an instance S (a finite set of covering families with covered$(f) \subseteq U$ and pc-flag $p(f)$) and a cell $c \in U$, the *signature* $\sigma_S(c)$ is

$$\begin{aligned} &\mathrm{H} : c \in P_{\mathrm{cov}}(S) \cap \mathrm{Hard6}, && \mathrm{C} : c \in P_{\mathrm{cov}}(S) \setminus \mathrm{Hard6}, \\ &\mathrm{Fh} : c \in (u(S) \setminus P_{\mathrm{cov}}(S)) \cap \mathrm{Hard6}, && \mathrm{Fo} : c \in (u(S) \setminus P_{\mathrm{cov}}(S)) \setminus \mathrm{Hard6}, \\ &\mathrm{Vh} : c \notin u(S),\ c \in \mathrm{Hard6}, && \mathrm{Vo} : c \notin u(S),\ c \notin \mathrm{Hard6}. \end{aligned} \tag{6}$$

$\mathrm{Spec}(S) = (\sigma_S(c))_{c \in U}$; the count vector is $(n_{\mathrm{H}}, n_{\mathrm{C}}, n_{\mathrm{Fh}}, n_{\mathrm{Fo}}, n_{\mathrm{Vh}}, n_{\mathrm{Vo}})$. The *forcing level* is

$$\psi_S(c) = \begin{cases} 2 & \sigma_S(c) \in \{\mathrm{H}, \mathrm{C}\}, \\ 1 & \sigma_S(c) \in \{\mathrm{Fh}, \mathrm{Fo}\}, \\ 0 & \sigma_S(c) \in \{\mathrm{Vh}, \mathrm{Vo}\}. \end{cases} \tag{7}$$

Assumption 1 (Cited census data)**.** The 78-group union census, the 3,081-pair-type spectrum census, and the 12,827-family covering census are supplied datums, each [COMPUTED] on two independent code paths with zero shared code and exact agreement [4, 8, 12]. All census numbers in this paper are cited data, not re-derived. The Lean theorems quantify over abstract (C, p) data, not over the census datasets.

3.2 Formalization and verification

The theorems of §4 are machine-checked in the repo `derekblair/math-proofs` (workspace checkout `~/workspace/math-unsolved/lean/`), modules `Mathproofs/ESAbsorptionOrder.lean`, `Mathproofs/ESForcingSpectrum.lean`, and `Mathproofs/ESForcingDecomp.lean` (each module header points back to this paper's directory). Verification config: Lean 4.34.1, Mathlib v4.34.1 (pinned commit `d13f23b`, Lake 5.0.0); bakkie `lake build` exit 0 (17,944 jobs, all succeeded) as part of the R322 34-theorem batch; zero `sorry`s; axioms [propext, Classical.choice, Quot.sound] standard for every theorem except `theoremC_filter`, which uses only [propext, Quot.sound] [12]. MD5-verified local↔bakkie matches; both modules registered 1× per side in `Mathproofs.lean` [12]. Rule-9 review gates: A1–A3 by Q-46 (SURVIVES), ρ/R3′/R3″/T/R4′ by Q-55 (SURVIVES), singleton factorization by Q-52 (SURVIVES), S2(a)′ by Q-56 (SURVIVES), S2(b–d)/S3/S4 by Q-53 (SURVIVES, with the original S2(a) objected to and repaired), A–D by Q-25 (SURVIVES conditional, four conditions in §4.5) [9, 10, 11, 12].

4 Results

4.1 The absorption poset

Theorem 1 (A1; PROVED)**.** *The absorption order* (3) *on union groups is a partial order:* ***absorp_refl***, ***absorp_trans***, ***absorp_antisymm***.

Theorem 2 (A2; PROVED)**.** *For union groups G, H, $\mathrm{datum}(G \cup H) = \mathrm{datum}(H) \iff G \preceq H$:* ***absorp_datum_iff***.

Theorem 3 (A3; PROVED)**.** *Let S be a finite set of families with union datum $D(S)$. If groups(S) contains $G \preceq H$ with $G \neq H$, removing all G-members leaves $D(S)$ unchanged:* ***dominance_prune***. *Hence any inclusion-minimal family set whose prime footprint covers a target $T \subseteq \mathbb{Z}/35$ has its groups forming a $\preceq$-antichain:* ***minimal_pcov_antichain***.

The A2 biconditional is computationally exact on the census: 0 violations over all 6,084 ordered pairs [COMPUTED]×2 [4].

A4 is [OPEN]. The surgical down-set statement (s surgical $\Rightarrow G \preceq s \iff C_G = \{c\}$) was hand-proved in the source dossier [4] but is *not* Lean-formalized and its rule-9 gate is unsatisfied; we make no claim about it here. The census datum behind it: 5 surgical groups (cells $5, 7, 12, 15, 22$), each with down-set of size exactly 2, and no surgical group on cell 0 [COMPUTED]×2 [4].

4.2 Redundancy rank

Let $\delta(S,T) := \rho(S) + \rho(T) - \rho(S \cup T)$ be the superadditivity deficit.

Proposition 1 (Naive R3 refuted; PROVED)**.** *The unstated-restriction claim $\rho(S\cup T) \geq \rho(S)+\rho(T)$ is false. Counterexample (Angel's, Q-48): with $4 \prec 3$ (census: $m_1 \prec \top$), $S = T = \{3,4\}$ gives $\rho(S \cup T) = 1 < 2 = \rho(S) + \rho(T)$, verified [COMPUTED]×2 [10].*

Theorem 4 (R3′ disjoint superadditivity; PROVED)**.** $S \cap T = \emptyset \Rightarrow \rho(S \cup T) \geq \rho(S) + \rho(T)$: *rho_superadd_disjoint*.

Theorem 5 (R3″ general corollary; PROVED)**.** *For all S, T: $\rho(S \cup T) \geq \rho(S) + \rho(T \setminus S)$:* *rho_superadd_general*.

Theorem 6 (T(iii) overlap-deficit bound; PROVED)**.** *Failure of naive superadditivity needs overlap (contrapositive of Theorem 4); overlap always realizes failure ($T := S$ with $\rho(S) > 0$); and for all S, T*

$$\delta(S,T) \leq \rho(T) - \rho(T \setminus S), \qquad \delta(S,T) \leq \rho(S) - \rho(S \setminus T) : \texttt{overlap_deficit_bound}. \tag{8}$$

The bound is one-sided: δ may be negative (strict superadditivity despite overlap; sampled minimum $\delta = -4$). It is sharp at $S = T = \{3,4\}$ ($\delta = 1 = 1 - 0$) [COMPUTED]×2 [10].

Theorem 7 (R4′ absorption-collapse; PROVED)**.** $G \prec H \Rightarrow \rho(\{G,H\}) = 1 = \rho(\{H\}) + 1$: *rho_absorption_collapse*, *with the reduction $\{G,H\} \to \{H\}$ unique:* `absorption_collapse_unique_core`.

The R4′ proof requires a case split: when $C_G = C_H$ (strictness forces $p_G = 0 < p_H = 1$), H is saved from redundancy by the pc-union, not by the cell inclusion [10]. All 303 strict census pairs give $\rho = 1$, including the 36 equal-cell pairs [COMPUTED]×2 [10]. Strictness of R3′ is genuine: $S = \{4,5\}$, $T = \{3\}$ (disjoint) give $\rho(S \cup T) = 2 > 0 + 0$ with unique core $\{\top\}$ [COMPUTED]×2 [10].

4.3 Singleton factorization

Among the 78 groups, $|C| \in \{1, 5, 7, 35\}$; 52 are singleton groups on 29 cells, and 6 cells $S_0 = \{0, 3, 4, 13, 18, 25\}$ carry no singleton group [7].

Theorem 8 (Singleton factorization; PROVED)**.** *Every minimal u-cover S of $\mathbb{Z}/35$ splits uniquely as $S = T \cup U$ where T is a $\preceq$-antichain of non-singleton groups, each with a T-private cell, with footprint $F(T) \supseteq S_0$, and U contains exactly one singleton group for each cell $c \notin F(T)$ (none on cells of $F(T)$); conversely every such pair is a minimal u-cover:* *singleton_factorization*, *singleton_factorization_converse*.

The census: 212,264 valid hard cores T (size distribution 3:24, 4:432, 5:3360, 6:14400, 7:38400, 8:63488, 9:62464, 10:29696) and exactly

$$\#\text{minimal } u\text{-covers of } \mathbb{Z}/35 = 7{,}052{,}450 \tag{9}$$

[COMPUTED]×2 (byte-identical certified stream digest, two fully independent enumerations; a third independent algorithm's direct enumeration did not complete in 15 minutes, itself a tractability measurement, compensated by an independent T-recount and a 2,000-cover direct-minimality audit) [7]. The two full-coverage singleton T's $\{(\mathbb{Z}/35, 0)\}$, $\{(\mathbb{Z}/35, 1)\}$ each contribute exactly one cover [7].

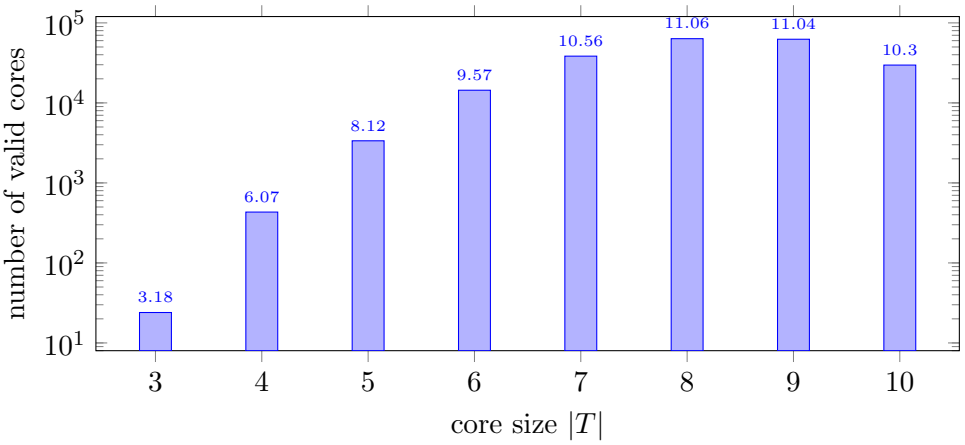


Figure 1: Hard-core size distribution of the 212,264 valid T's in Theorem 8 [COMPUTED]×2, log scale. The 7,052,450 minimal u-covers of (9) are the per-T singleton-fill products summed over these cores.

4.4 The forcing spectrum

The sum identities are the spectrum's consistency core: $n_{\mathrm{H}} = m$, $n_{\mathrm{C}} = C$, $n_{\mathrm{Fh}} + n_{\mathrm{Fo}} = F$, $n_{\mathrm{Vh}} + n_{\mathrm{Vo}} = 35 - \mathrm{cap}_2$, $n_{\mathrm{H}} + n_{\mathrm{C}} = \mathrm{cap}_1$, $n_{\mathrm{C}} + n_{\mathrm{Fh}} + n_{\mathrm{Fo}} = T$ (S1 [8], definitional unfolding).

Theorem 9 (S2(a)′ repaired forced-cell characterization; PROVED)**.** *For $c \in U$:*

$$\sigma_S(c) \in \{\mathrm{Fh}, \mathrm{Fo}\} \iff c \in u(S) \wedge (\forall f \in S)(c \in \mathrm{covered}(f) \Rightarrow \neg p(f)) : \texttt{s2a'_forced}. \tag{10}$$

Theorem 10 (S2(b–d); PROVED)**.** *For $c \in U$: $\sigma_S(c) \in \{\mathrm{H}, \mathrm{C}\} \iff (\exists f \in S)(p(f) \wedge c \in \mathrm{covered}(f))$ (pc-secured: `s2b_pc`); $\sigma_S(c) \in \{\mathrm{Vh}, \mathrm{Vo}\} \iff c \notin u(S)$ (`s2c_void`); the hard/open split is exactly the $c \in$ Hard6 check (`s2d_split`).*

Theorem 11 (S3 partition; PROVED)**.** *The six classes (6) partition U (pairwise disjoint, exhaustive): `spectrum_partition`. The ambiguous middle — a cell covered both by a pc and by a composite-only family — is resolved by the priority rule: pc-secured wins.*

Theorem 12 (S4 union monotonicity; PROVED)**.** *With ψ from (7),*

$$\psi_{S \cup T}(c) = \max(\psi_S(c), \psi_T(c)) : \texttt{psi_union_max}. \tag{11}$$

Upgrades: $\mathrm{Vh} \to \mathrm{Fh}/\mathrm{H}$, $\mathrm{Vo} \to \mathrm{Fo}/\mathrm{C}$, $\mathrm{Fh} \to \mathrm{H}$, $\mathrm{Fo} \to \mathrm{C}$; $\{\mathrm{H}, \mathrm{C}\}$ *absorbing.*

Honest repair history. S2(a) *as originally stated* is false: its right-hand side holds vacuously at every uncovered cell, misclassifying all void cells as slack-forced (hostile review Q-53 [11]). The repair S2(a)′ restores the dropped conjunct $c \in u(S)$ (10); the failure set of the original is exactly the void cells — 88,259 (unweighted) pair-type×cell checks fail, all at $c \notin u(S)$, zero at covered cells [COMPUTED]×2 [11]. S2(b–d), S3, S4, and the C1–C4 corollaries survived Q-53 unchanged [11].

The census (both paths, 3,081 pair types, $\sum w = 82{,}272{,}378$): aggregate masses

$$\mathrm{H}\,5.27\%,\ \mathrm{C}\,26.66\%,\ \mathrm{Fh}\,6.41\%,\ \mathrm{Fo}\,35.15\%,\ \mathrm{Vh}\,5.46\%,\ \mathrm{Vo}\,21.05\%, \tag{12}$$

with 15.43% of forced slack on hard cells (Fh/(Fh + Fo)) — hard cells whose only coverage is composite-only, invisible to the aggregate F. Upgrade anatomy over pair types: 18,948 cell upgrades (Vh→H 1616, Vh→Fh 1616, Vo→C 7682, Vo→Fo 7540, Fh→H 89, Fo→C 405): $F \to$pc "rescues" are rare (494) [8, 11]. Cell-level: cell 0 has the lowest H mass of the six hard cells (30.31% vs 30.75% hard mean, quantifying the no-surgical-group gap); cell 16 is covered in 92.3% of pairs but pc-secured in only 30.27% [8, 11]. All [COMPUTED]×2. (Terminology note: "forcing spectrum" denotes something unrelated in matching-forcing theory [3]; the qualified name *ES* forcing spectrum avoids the collision [8].)

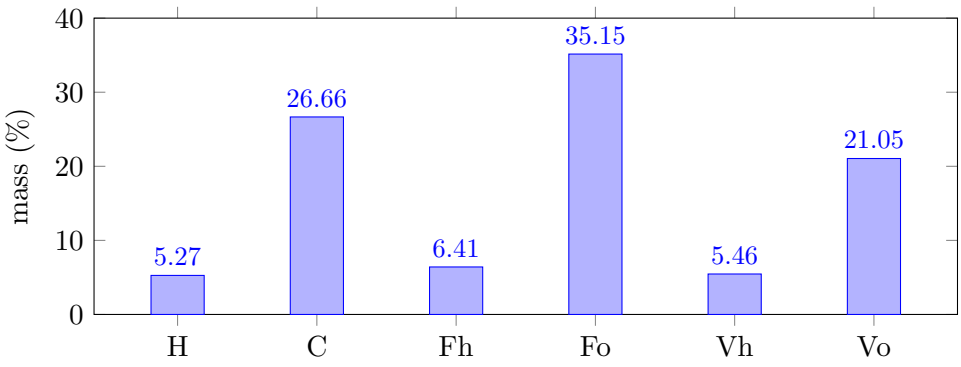


Figure 2: Aggregate six-class masses (12) of the ES forcing spectrum over the pair census (3,081 pair types, $\sum w = 82{,}272{,}378$) [COMPUTED]×2. The spectrum strictly refines (T, F, C): 35 of 48 distinct (T, F, C) triples admit ≥ 2 distinct spectra [8].

4.5 The forcing decomposition A–D

The two-sided forcing decomposition of the R281 census [12] is formalized as [PROVED]:

- **Theorem A** (open–hard coincidence): $n = 24k+1 \equiv H \pmod{840} \iff k \equiv (H-1)/24 \pmod{35}$ for the six hard classes (`theoremA_crt`, `decide`); the six k-classes are exactly $k \bmod 7 \in \{0,1,5\} \times k \bmod 5 \in \{0,2\}$ (`theoremA_crt_box`).
- **Theorem B** (scaling-forced composites): a solution of (1) for n scales to one for kn (`theoremB_scaling`, field algebra).
- **Theorem C** (filter = F-axis): definitional unfolding (`theoremC_filter`).
- **Theorem D** (surgical characterization): divisor analysis plus `decide` (`theoremD_card`, `theoremD_mod7`, `theoremD_surgical`).

The Q-25 review verdict (SURVIVES conditional [12]) binds the following conditions, stated verbatim from the module docstrings: (1) B's Reading carries the prime-free-class lemma: every non-prime-covering class is prime-free, i.e. $\gcd(24k_0+1, 24 \cdot \text{mprime}) > 1$ — certified 8,965/8,965 on the census data, NOT proved in general. B alone does not make F forced. (2) 'prime-covering' := R7's $x \cdot h$ criterion. The 'C = prime coverage' glosses are dropped: 262 pc families are prime-free, and 38.2% of C units lie in cell sub-progressions with no primes. (3) Aggregate slack is stated as unweighted cell hits (m counts cells hit, not density; the density-weighted F share is 55.0%, not 68.3%). (4) R7 tests $ab \mid x \cdot h$, NOT $\text{lcm}(a,b) \mid x \cdot h$ (5,582 families pass the lcm test; the 1,720 extra are all prime-free).

5 Discussion

The census poset's shape. The 78 groups are [COMPUTED]×2 data [4]: 303 strict absorption pairs; unique maximal element $\top = (\mathbb{Z}/35, 1)$ with 1,945 families (every group $\preceq \top$); 29 minimal elements (25 composite-only singletons plus 4 prime-covering singletons on non-hard cells); height 4, width 39 (Table 2, Figure 3). Deeper structure [5]: rank distribution 29/35/13/1 (ranks 0–3), 164 covers with 8 "jump" covers (rank-0⋖rank-2), hence the poset is *not* graded. The census poset is a join-semilattice (6,084/6,084 joins; abstract join in census only 870/6,084) but not a lattice (1,054/6,084 meets) [5]. Minimal Hard6 prime-covers number exactly 16, all antichains (A3 audit); surgical-only coverage is impossible since cell 0 carries no singleton [5]. The 8.9×10^{13} antichains (single-path count) collapse to 212,264 hard cores under the singleton factorization of Theorem 8 — the count (9) closes the open Z35 minimal-u-cover enumeration [5, 7].

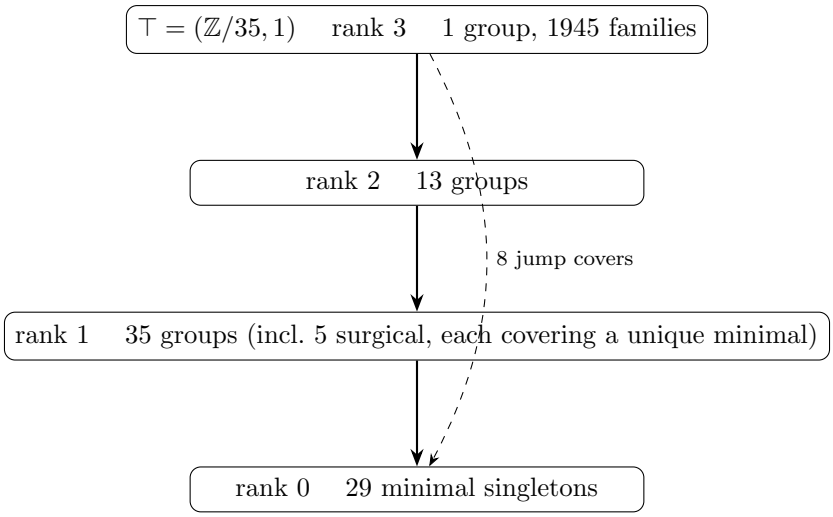


Figure 3: Schematic of the census absorption poset (not to scale): counts are exact [COMPUTED]×2 [4, 5]. Every group absorbs into the unique top $\top$; surgical rank-1 groups sit atop their unique minimal cell-mates (A4's down-set is the [OPEN] item of §4.1).

The greedoid episode, honestly labeled. Beyond the formalized core, the program developed a greedoid-level theory on the poset [6]: both candidate feasible families satisfy accessibility (D1); exchange fails for both, witnessed by $X = \{4, 5\}$, $Y = \{\top\}$ (D2); the failure is exactly characterized by the comparability shadow (D3), also failing with no top in sight ($X = \{4, 6\}$, $Y = \{0\}$, D3′); and the redundancy rank refutes core canonicity: $S = \{0, 4, 6, 17, 22, 29, 35, 38\}$ has exactly two cores, $\{0\}$ and the seven singletons, so $\rho(S) = 7$ (R5). These all SURVIVED hostile review Q-48 [10] but are *not Lean-formalized*: we report them as [PROVED*] context, not as claimed theorems of this paper.

Lessons from the two objections. The Q-48 objection [10] showed the naive R3 proof transferred a reduction sequence across the $S \cap T \neq \emptyset$ boundary where it cannot transfer; the disjoint restriction is not a weakening of convenience but the true statement, and the general corollary R3″ plus the deficit bound (8) say exactly how much overlap can destroy. The Q-53 objection [11] showed a dropped conjunct ($c \in u(S)$) silently universalized a characterization to cells where its hypothesis was vacuous. Both repairs survived re-review (Q-55, Q-56 [12]).

Limitations. (i) The Lean theorems quantify over abstract (C, p) data; transfer to the 78-group census rests on the reconstruction fidelity of Assumption 1, not on a machine-checked census. (ii) The spectrum census (12) describes the R281 census universe, not general ES theorems [11]. (iii) The deficit bound is one-sided; no exact $\delta(S, T)$ formula is known, and maximal exchange-closed subfamilies are open [10, 6]. (iv) The monotone R59/R63 audit of all proofs is a human audit, not a machine check. (v) A4 is [OPEN]; minimal-cover enumeration beyond the group level (Q-30's family-level machinery) is a disjoint layer, not pursued here [7].

6 Conclusion

We have given the union layer of the Erdős–Straus covering program an order theory and a cellwise forcing profile, both machine-checked. The absorption order $\preceq$ of (3) is a partial order (Theorem 1) that characterizes datum-equality (Theorem 2) and prunes dominated families while preserving the datum (Theorem 3). The redundancy rank ρ of (5) satisfies disjoint superadditivity (Theorem 4), its general corollary (Theorem 5), the one-sided overlap-deficit bound

(Theorem 6), and the absorption-collapse formula with uniqueness (Theorem 7) — with the naive unrestricted superadditivity honestly refuted (Proposition 1). The singleton factorization (Theorem 8) reduces the minimal u-cover census of $\mathbb{Z}/35$ to a 2^{24} factorization yielding the exact count 7,052,450. The ES forcing spectrum (6) carries the repaired forced-cell characterization (Theorem 9), the pc-secured/void/split laws (Theorem 10), the six-class partition (Theorem 11), and the union-max rule (Theorem 12), together with its complete pair-type census. Theorems A–D close the cited two-sided decomposition under four documented conditions. All of this is [COMPUTED]×2 where data and [PROVED] where mathematics, except the surgical down-set A4, which we leave explicitly [OPEN].

7 References

References

[1] P. Erdős, *On a Diophantine equation*, Mat. Lapok **1** (1950), 192–210. (The Erdős–Straus conjecture.)

[2] L. J. Mordell, *Diophantine Equations*, Academic Press, 1969. (Theorem A's CRT computation; Mordell's list of admissible squares mod 840.)

[3] P. Adams, M. Mahdian, and E. S. Mahmoodian, *On the forced matching numbers of bipartite graphs*, Discrete Math. **281** (2004), 1–12. ("Forcing spectrum" in matching-forcing theory is an unrelated object; our qualified name *ES* forcing spectrum avoids the collision.)

[4] D. Blair, *Unsolved Math — Round 304: the ES absorption poset*, program dossier, 2026-10-11. (Supplied datum: 78-group census; A1–A4, Q-46 SURVIVES.)

[5] D. Blair, *Unsolved Math — Round 308: absorption-poset antichains, redundancy & lattice structure*, program dossier, 2026-10-11. (Supplied datum: B1/B2, join-semilattice, minimal Hard6 covers.)

[6] D. Blair, *Unsolved Math — Round 311: the cover greedoid of the ES absorption poset*, program dossier, 2026-10-11. ([PROVED*] D1–D3, R1–R5; Q-48 SURVIVES for all but R3/R4's wording.)

[7] D. Blair, *Unsolved Math — Round 312: Z35 minimal u-cover census via singleton-factorization*, program dossier, 2026-10-11. (Supplied datum: 212,264 cores, 7,052,450 covers [COMPUTED]×2; Q-52 SURVIVES.)

[8] D. Blair, *Unsolved Math — Round 313: ES forcing spectrum*, program dossier, 2026-10-11. (Supplied datum: spectrum census 3,081 pair types; S1–S4/C1–C4, Q-53 SURVIVES except S2(a) as stated.)

[9] D. Blair, *Unsolved Math — Round 314: Lean + rule-9 lane*, program dossier, 2026-10-11. (Supplied datum: Q-46 quoted SURVIVES verdict.)

[10] D. Blair, *Unsolved Math — Round 315: repair of the cover-greedoid theory after Q-48*, program dossier, 2026-10-11. (Supplied datum: R3 refutation, R3′/R3″/T/R4′ repairs, Q-48 quoted OBJECTION verbatim.)

[11] D. Blair, *Unsolved Math — Round 316: ES forcing-spectrum S2(a) repair (after Q-53 OBJECTION)*, program dossier, 2026-10-11. (Supplied datum: S2(a)′ repair, 88,259-check failure census, Q-53 quoted OBJECTION verbatim.)

[12] D. Blair, *Unsolved Math — Round 322: Lean + rule-9 lane (fold complete)*, program dossier, 2026-10-11. (Supplied datum: 34-theorem bakkie build — exit 0, zero sorrys, standard axioms; Q-25 SURVIVES conditional with four conditions; Q-55 and Q-56 quoted SURVIVES verdicts.)

Table 1: Lean theorem inventory. Every theorem below is [PROVED] under the verification config of §3.2. Module paths are relative to the repo root `derekblair/math-proofs`.

module	claim	dossier / gate	Lean theorem(s)
`Mathproofs/ESAbsorptionOrder.lean`	A1 partial order	R304 / Q-46	absorp_refl, absorp_trans, absorp_antisymm
	A2 datum characterization	R304 / Q-46	absorp_datum_iff
	A3 pruning + antichain corollary	R304 / Q-46	dominance_prune, minimal_pcov_antichain
	R2 redundancy monotonicity	R308 / Q-48	redundant_mono
	R3′/R3″/T(iii)	R315 / Q-55	rho_superadd_disjoint, rho_superadd_general, overlap_deficit_bound
	R4′ collapse + uniqueness	R315 / Q-55	rho_absorption_collapse, absorption_collapse_unique_core
	singleton factorization ($\Longleftrightarrow$)	R312 / Q-52	singleton_factorization, singleton_factorization_converse
`Mathproofs/ESForcingSpectrum.lean`	S2(a) unpaired	R316 / Q-56	`s2a'_forced`
	S2(b–d)	R313 / Q-53	s2b_pc, s2c_void, s2d_split
	S3 partition	R313 / Q-53	spectrum_partition
	S4 union max	R313 / Q-53	psi_union_max
`Mathproofs/ESForcingDeficit.lean`	A CRT box	R281 / Q-25	theoremA_crt, theoremA_crt_box
	B scaling	R281 / Q-25	theoremB_scaling
	C filter	R281 / Q-25	theoremC_filter
	D card/mod7/surgical	R281 / Q-25	theoremD_card, theoremD_mod7, theoremD_surgical

Table 2: Selected census facts [COMPUTED]×2 [4, 5, 7, 8, 11].

quantity	value
union groups	78
strict absorption pairs	303
top group $(\mathbb{Z}/35, 1)$ families	1,945
minimal elements	29
height / width	4 / 39
covers (rank-0⋖rank-2 jumps)	164 (8)
census joins / meets	6,084 / 1,054
surgical groups (cells $5, 7, 12, 15, 22$)	5, down-set size 2 each
valid hard cores T	212,264
minimal u-covers of $\mathbb{Z}/35$	7,052,450
singleton groups / singleton-cells	52 / 29
spectrum pair types / weighted pairs	3,081 / 82,272,378